Geometry Review: Unit 3 — Triangles & Congruence

Station 1 of 4: Classifying Triangles and the Triangle Inequality

Name: _____ Date: _____

Answer each question. Show your reasoning for the triangle inequality problems.

1. Classify a triangle with side lengths $5, 5, 8$ by its sides.

2. Classify a triangle with side lengths $7, 9, 12$ by its sides.

3. Classify each triangle by its angles: (a) $30°, 60°, 90°$ (b) $40°, 70°, 70°$ (c) $25°, 30°, 125°$.

4. Can sides of lengths $4, 7, 10$ form a triangle? Explain.

5. Can sides of lengths $3, 5, 9$ form a triangle? Explain.

6. Two sides of a triangle are 8 and 13. Find the range of possible lengths of the third side x.

7. Two sides of a triangle are both 6. Find the range of possible lengths of the third side x.

8. An isosceles triangle has two side lengths 4 and 9. What is its perimeter?

9. In $\triangle ABC$, $m\angle A = 50°$, $m\angle B = 60°$, $m\angle C = 70°$. List the sides from shortest to longest.

10. Find the measure of each angle of an equilateral triangle.

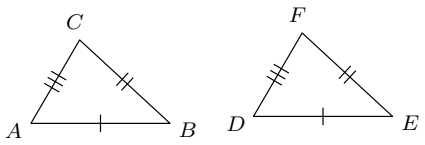
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Station 2 of 4: Proving Triangles Congruent

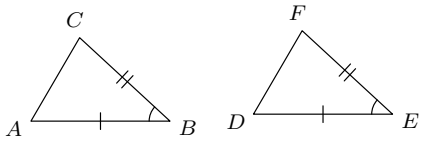
Name: _____ Date: _____

Name the congruence shortcut (SSS, SAS, ASA, AAS, HL) that justifies each statement, or write "not enough information." Diagrams are not drawn to scale.

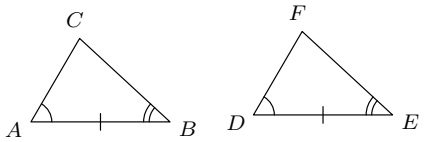
1. Are the triangles congruent? If so, name the shortcut; if not, write "not enough information."



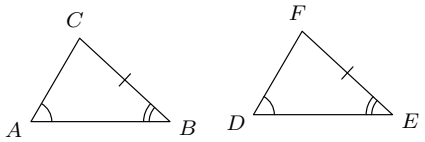
2. Are the triangles congruent? If so, name the shortcut; if not, write "not enough information."



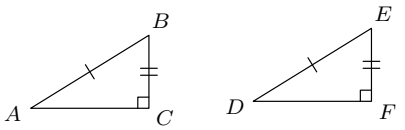
3. Are the triangles congruent? If so, name the shortcut; if not, write "not enough information."



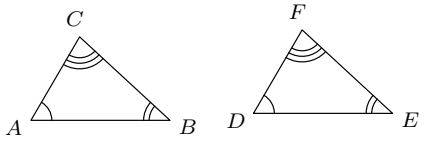
4. Are the triangles congruent? If so, name the shortcut; if not, write "not enough information."



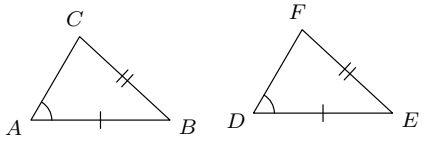
5. Are the triangles congruent? If so, name the shortcut; if not, write "not enough information."



6. Are the triangles congruent? If so, name the shortcut; if not, write "not enough information."



7. Are the triangles congruent? If so, name the shortcut; if not, write "not enough information."



8. If $\triangle ABC \cong \triangle PQR$, which side is congruent to $\overline{BC}$? Which angle is congruent to $\angle C$?

9. $\triangle ABC \cong \triangle PQR$, $AB = 3x + 1$, and $PQ = 19$. Find x.

10. Given $\overline{AB} \cong \overline{DE}$, $\angle B \cong \angle E$, and $\overline{BC} \cong \overline{EF}$, which shortcut proves $\triangle ABC \cong \triangle DEF$? What does CPCTC then tell you about $\overline{AC}$?

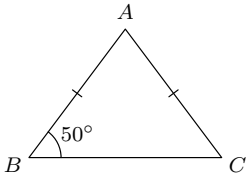
Geometry Review: Unit 3 — Triangles & Congruence

Station 3 of 4: Isosceles and Equilateral Triangles

Name: _____ Date: _____

Use the Isosceles Triangle Theorem and its converse. Show your equations. Diagrams are not drawn to scale.

1. In isosceles $\triangle ABC$ with $AB = AC$, $m\angle B = 50°$. Find $m\angle C$ and $m\angle A$.



2. An isosceles triangle has a vertex angle of 40°. Find each base angle.

3. The base angles of an isosceles triangle measure $(3x + 5)°$ and $(5x - 15)°$. Find x, the base angles, and the vertex angle.

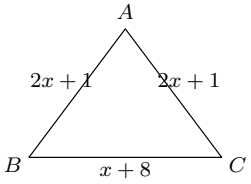
4. An equilateral triangle has sides $(3x + 2)$ and $(5x - 6)$. Find x and the perimeter.

5. In $\triangle ABC$, $AB = AC$. If $AB = 7x - 4$ and $AC = 4x + 11$, find x and AB.

6. A triangle has angles 70°, 55°, and 55°. Which two sides are congruent?

7. The base angles of an isosceles triangle are $\angle B = (2x + 10)°$ and $\angle C = (4x - 20)°$. Find x and the vertex angle $\angle A$.

8. An isosceles triangle has legs of length $(2x + 1)$ and a base of length $(x + 8)$. If the perimeter is 50, find x and all three side lengths.



9. The perimeter of an equilateral triangle is 51. Find the length of each side.

10. Each leg of an isosceles right triangle is 6. Find the measure of each acute angle.

Geometry Review: Unit 3 — Triangles & Congruence

Station 4 of 4: Midsegments, Medians, and Special Segments

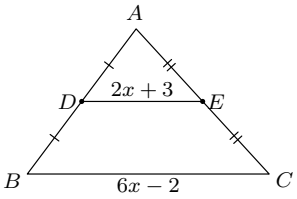
Name: _____ Date: _____

Apply the Triangle Midsegment Theorem and properties of the special segments in triangles. Diagrams are not drawn to scale.

1. A midsegment of a triangle is parallel to a side of length 14. How long is the midsegment?

2. A midsegment of a triangle has length 9. How long is the side parallel to it?

3. In $\triangle ABC$, D and E are the midpoints of $\overline{AB}$ and $\overline{AC}$. If $DE = 2x + 3$ and $BC = 6x - 2$, find x, DE, and BC.



4. A triangle has sides $10, 14, 16$. Find the perimeter of the triangle formed by connecting the midpoints of its sides.

5. The centroid divides each median in a $2 : 1$ ratio. A median is 18 long. How far is the centroid from the vertex? From the midpoint of the opposite side?

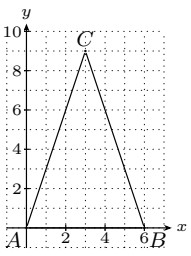
6. The centroid is 8 units from the midpoint of a side along a median. Find the length of the whole median.

7. P is on the perpendicular bisector of $\overline{AB}$. If $PA = 3x + 5$ and $PB = 5x - 7$, find x and PA.

8. A point on the bisector of an angle is equidistant from the sides. If the distances are $(4x - 3)$ and $(2x + 9)$, find x and the distance.

9. Name the point of concurrency of: (a) the three medians (b) the three altitudes (c) the three perpendicular bisectors (d) the three angle bisectors.

10. Find the centroid of the triangle with vertices $A(0, 0)$, $B(6, 0)$, and $C(3, 9)$.



Geometry Review: Unit 3 — Triangles & Congruence

Answer Key

Station 1: Classifying Triangles and the Triangle Inequality

1. Isosceles
2. Scalene
3. (a) right; (b) acute; (c) obtuse
4. Yes; $4 + 7 = 11 > 10$ (and the other sums are larger)
5. No; $3 + 5 = 8 < 9$
6. $5 < x < 21$
7. $0 < x < 12$
8. 22 *sides* $9, 9, 4$ *(since* $4 + 4 < 9$*)*
9. $BC < AC < AB$
10. $60°$

Station 2: Proving Triangles Congruent

1. SSS
2. SAS
3. ASA
4. AAS
5. HL
6. Not enough information (AAA proves similarity, not congruence)
7. Not enough information (SSA does not work)
8. $\overline{QR}$; $\angle R$
9. $x = 6$
10. SAS; $\overline{AC} \cong \overline{DF}$

Station 3: Isosceles and Equilateral Triangles

1. $m\angle C = 50°,\ m\angle A = 80°$
2. $70°$ $\quad \frac{180 - 40}{2}$
3. $x = 10;\ 35°$ each; $110°$
4. $x = 4;\ P = 42$
5. $x = 5;\ AB = 31$
6. The two sides opposite the $55°$ angles
7. $x = 15;\ m\angle A = 100°$
8. $x = 8;\ 17,\ 17,\ 16$
9. 17
10. $45°$

Station 4: Midsegments, Medians, and Special Segments

1. 7
2. 18
3. $x = 4;\ DE = 11,\ BC = 22$
4. 20
5. 12; 6
6. 24
7. $x = 6;\ PA = 23$
8. $x = 6;\ 21$
9. centroid; orthocenter; circumcenter; incenter
10. $(3, 3)$ *average the coordinates*